From Pedagogical Siamese Metric Learning to Falsifiable Biometric Embedding Compression

A Replayable Research and Engineering Programme

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Independent exploratory research
https://github.com/gharbonnier78/siamese-embedding-compression-lab

Version 0.2.0-draft — August 2026

Scientific status. This manuscript is a research protocol and an honest report of a completed preliminary study. It is not peer reviewed, not an industrial biometric evaluation, and not evidence of regulatory conformity. Planned studies contain no experimental results.

Abstract

Dimensionality reduction is attractive for large biometric galleries because template payload affects storage, memory traffic, cache behaviour and search cost. Yet the ability to compute a cosine distance after projection is not evidence that the projected template preserves decision-relevant performance, nor that supervised metric learning adds value over ordinary compression. This work converts an educational Siamese-network exercise into a falsifiable, replayable research and engineering programme. The programme separates mechanism validity, representation quality, threshold transfer, operational value and claim admissibility. It predeclares a non-inferiority estimand at a target false match rate; requires raw, random and principal-component controls; enforces identity-disjoint development, validation-frozen thresholds and a once-opened test set; and attaches every claim to automated evidence gates. A completed preliminary study uses frozen ImageNet ResNet-18 features on the Labeled Faces in the Wild development protocol. The contrastive training mechanism is real, but non-inferiority of the 128-dimensional Siamese projection to the raw 512-dimensional representation is not demonstrated. The result is intentionally retained as negative evidence. Future registered studies replace the unsuitable extractor, strengthen dataset representativity, add external and shifted evaluation, and measure 1:N retrieval and end-to-end engineering cost. Immutable MMALS replay bundles preserve the execution history. MMALS-CAL provides a three-state internal claim gate—admissible, inadmissible or indeterminate—without being presented as a complete safety proof.

1 Research position

The object under study is not "face recognition by Siamese networks" in the broad sense. It is a bounded intervention applied to an already trained extractor. A frozen extractor f maps an image x to a source embedding $z = f(x) \in \mathbb{R}^D$. A projection g_θ maps z to $u = g_\theta(z) \in \mathbb{R}^d$, where $d < D$. The matcher compares two projected templates or searches a gallery built from them. Training and inference are therefore distinct:

$$\text{training:} \qquad \theta^* = \arg\min_\theta \sum_{(i,j)} \mathcal{L}\big(g_\theta(z_i), g_\theta(z_j), y_{ij}\big), \tag{1}$$

$$\text{inference:} \qquad s(x_i, x_j) = \operatorname{sim}\big(g_{\theta^*}(f(x_i)), g_{\theta^*}(f(x_j))\big). \tag{2}$$

For a contrastive objective and distance d_{ij}, a representative loss is

$$\mathcal{L}_{ij} = y_{ij} d_{ij}^2 + (1 - y_{ij}) \max(0, m - d_{ij})^2, \tag{3}$$

where $y_{ij} = 1$ denotes the same identity and m is a margin. The two branches share weights. Pair labels do not merely calibrate a threshold: their gradient changes the projection and therefore the geometry of the embedding space.

The programme asks whether this learned geometry is useful after controlling for dimension, data, extractor, operating point and system implementation. A positive answer must survive comparisons against the uncompressed representation and matched compression controls. A negative answer is a valid result.

1.1 Why this is not closed-set identity classification

A classifier head can be used while training a face extractor, with training identities acting as classes. Production recognition is normally open set: enrolled identities can be different from training identities, and decisions are produced from similarities, thresholds or ranked candidates. Modern margin-based classification objectives such as ArcFace learn a discriminative representation during training, while verification and identification consume embeddings at inference [4]. Siamese pair training is another supervision route; it is not required for every deployed facial matcher and is not synonymous with one-shot operation.

2 Research questions and falsifiable claims

The umbrella question is:

> Can supervised metric projection reduce biometric template and system cost while preserving decision-relevant verification and identification performance relative to the uncompressed embedding and matched compression controls?

The claim registry is machine-readable in `claims/registry.yaml`. Table 1 summarises the current state. “Open” is not weak positive evidence; it means that the claim has not been tested by an admissible study.

Table 1: Claim register at programme version 0.2.

ID	Bounded claim	Current status
`C-MECHANISM-001`	Pair supervision updates the shared projection	Established, limited
`C-STORAGE-001`	Float32 payload falls from 512D to 128D by factor four	Arithmetic only
`C-NI-001`	Siamese-128 is non-inferior to raw-512 in Study 0	Not demonstrated
`C-SUP-001`	Pair supervision adds value over PCA/random	Not demonstrated
`C-FACE-001`	Result transfers to face-specific backbones/external data	Open
`C-1N-001`	Compression preserves 1:N and improves system cost	Open
`C-CAL-001`	CAL rejects or abstains on unsupported claims	Specified, not qualified

2.1 Primary representation hypothesis

Let b be the raw representation, m a candidate route, and α the target FMR. At separately located equal-FMR benchmark points,

$$\Delta_{\mathrm{FNMR}}(m, \alpha) = \mathrm{FNMR}_m(\alpha) - \mathrm{FNMR}_b(\alpha). \tag{4}$$

Given a predeclared non-inferiority margin δ, the candidate is non-inferior only if

$$\mathrm{UCB}_{1-\gamma}\left[\Delta_{\mathrm{FNMR}}(m, \alpha)\right] \leq \delta, \tag{5}$$

for every predeclared seed under the Study 0 robustness rule. The benchmark threshold may use test labels solely to compare discrimination at the same empirical FMR; it is marked non-deployable. A separate operational analysis selects the threshold on VALIDATION, freezes it, and applies it once to TEST.

2.2 Supervision-value hypothesis

Non-inferiority to raw does not establish value added by pair supervision. At equal output dimension and matched inputs, the Siamese route must also be compared with random projection and train-only PCA. Study 2 will predeclare either a superiority margin or a Pareto rule. It will not select the metric after seeing test results.

2.3 Engineering hypothesis

The arithmetic reduction from D to d is not the engineering estimand. Let total system cost under workload w be

$$C_m(w) = C_{\text{extract}} + C_{\text{project}} + C_{\text{store}} + C_{\text{index}} + C_{\text{search}} + C_{\text{replicate}}. \tag{6}$$

An engineering claim requires a predeclared reduction in one or more resources while all mandatory accuracy, latency and reliability constraints remain satisfied. Extractor weights, projection weights, metadata, encryption, index overhead and replicas must not be hidden.

3 Programme architecture

The programme is incremental. Each study has a maximum permitted conclusion and later studies do not rewrite earlier evidence.

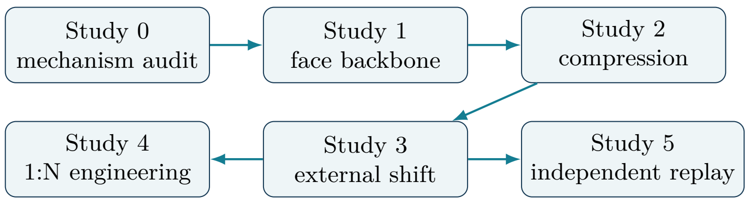

Figure 1: Bounded studies. A gate failure lowers the claim level and can stop expansion.

Study 0 — mechanism audit. Completed frozen ImageNet ResNet-18/LFW experiment. Its maximum conclusion is limited to the mechanism and declared protocol.
Study 1 — face-specific representation. Replace the unsuitable source extractor, pin weights and provenance, and test the primary 1:1 non-inferiority claim.
Study 2 — compression ablation. Compare dimensions, projection families and quantisation under matched budgets and a multiplicity policy.
Study 3 — external validity. Evaluate untouched datasets and predeclared capture regimes, including shifts in quality, pose, time interval and sensor where authorised.
Study 4 — 1:N engineering. Separate representation error from exact or approximate index behaviour, and measure accuracy, RAM, storage, latency and throughput jointly.
Study 5 — independent reproduction. Replay an immutable bundle in a distinct environment or by a distinct team and report discrepancies rather than hiding them.

4 Dataset requirements and representativity

ISO/IEC 19795-1:2021 frames biometric performance testing in terms of specified metrics, test protocols, data records, bias reduction and explicit limits of applicability [6]. This programme pins that published edition as its normative snapshot; an edition under development is not treated as an adopted requirement.

Every qualification dataset requires a versioned datasheet. The datasheet must state:

- provenance, licence, acquisition mechanism and cryptographic digests;
- target use case, target population and populations not covered;
- identities, captures per identity, sensors and known capture conditions;

- inclusion, exclusion, missingness and quality rules;
- distributions of pose, occlusion, age interval and relevant regimes when authorised;
- split unit, duplicate audit, cross-dataset overlap and contamination controls;
- construction of genuine/impostor comparisons or mated/non-mated searches;
- effective statistical resolution at the target operating point;
- privacy, security, retention and redistribution constraints;
- allowed and forbidden conclusions.

Public availability is not representativity. A web-photo benchmark, a controlled identity document capture and a border kiosk can induce different quality and error distributions. The study must define the intended inference domain before using the word representative.

4.1 Splits and dependence

Identity is the default split unit. No identity may cross development and qualification. Near duplicates and derived images require explicit audit. Because comparisons reuse identities and images, pair observations are not automatically independent. Uncertainty must be resampled at an identity or justified cluster level. A naive pair-level confidence interval may be reported only as a diagnostic alongside the dependency-aware estimator.

4.2 Operating-point resolution

An empirical set with N_I impostor comparisons has step size $1/N_I$. With zero observed errors, the approximate rule-of-three upper 95% bound $3/N_I$ is a sanity check, not a complete power analysis and not a correction for dependence. Before qualification, the protocol must predeclare either desired precision for FMR/FNMR or power for the paired non-inferiority estimand, then demonstrate that the effective sample supports it.

5 Experimental protocol

5.1 Common input and controlled routes

Every route receives exactly the same frozen source embeddings and split assignments. The mandatory routes are:

1. raw L2-normalised D-dimensional embedding;
2. seeded Gaussian projection to d dimensions without learning;
3. PCA fitted only on TRAIN endpoints and projected to d dimensions;
4. shared learned projection trained from labelled TRAIN pairs.

Study 2 may add nonlinear heads, autoencoders or quantised routes, but never remove the mandatory controls. Output dimension and representation normalisation must be matched.

5.2 Execution order

The enforced logical order is:

1. freeze configuration, implementation, data manifests, claims and seeds;
2. validate identity-disjoint TRAIN, VALIDATION and TEST;
3. fit projections on TRAIN only;
4. early-stop and select hyperparameters on VALIDATION only;
5. select each operational threshold on VALIDATION only;
6. freeze every route and operational threshold;
7. open TEST exactly once;
8. report validation-frozen operating metrics;
9. compute non-deployable equal-FMR benchmark points for representation comparison;
10. report all declared seeds and all mandatory controls;
11. evaluate gates and generate the bounded claim statement.

Any violation is preserved in the audit trace. Re-running after test inspection creates a new exploratory study identifier; it cannot retain confirmatory status.

5.3 Statistical analysis

The primary analysis is paired because candidate and reference are evaluated on the same identities. Bootstrap replicates preserve the label and identity structure. Confidence level, number of replicates, random seed, margin and operating point are configuration fields. Secondary EER and ROC AUC estimates are descriptive unless separately registered.

For multiple dimensions, losses or backbones, the study must predeclare a family of tests and either control family-wise error, control false discovery rate, or label the analysis exploratory with simultaneous intervals where suitable. Reporting only the best seed or best dimension on TEST is prohibited.

6 Engineering evaluation

Verification and identification are separate studies. NIST FRTE reports 1:1 FNMR at specified FMRs and evaluates 1:N using FNIR/FPIR, gallery size and ranked candidates; it also reports template generation, search duration and resource measures [7, 8]. This programme adopts the distinction and terminology, not NIST's sequestered datasets or the authority of an FRTE result.

Study 4 will report at least:

- 1:1 FNMR/FMR with validation-frozen and descriptive equal-FMR views separated;
- 1:N FNIR/FPIR and Recall@K for mated and non-mated searches;
- gallery size, templates per identity and probe regime;
- exact search and each approximate-nearest-neighbour configuration separately;
- template payload, metadata, index RAM, persistent storage and replication;
- extractor, projection, index build and search durations;
- cold/warm cache, concurrency, throughput and tail latency;
- hardware, software, thread count and energy method when energy is claimed.

The expected scientific product is a Pareto frontier, not necessarily one universal winner. A 128D representation may be useful for a large in-memory watchlist and irrelevant for a phone holding one or two enrolled templates.

7 Scientific gates and continuous integration

Continuous integration verifies evidence contracts; it does not convert code quality into biometric validity. Gates have four states: PASS, FAIL, INDETERMINATE and NOT RUN.

Table 2: Research qualification gates.

Gate	Purpose	Examples of required evidence
G0	Software and contract integrity	Schemas, gradients, deterministic smoke replay, hash verification
G1	Experimental design	Identity separation, train-only fitting, freeze-before-test, all seeds
G2	Statistical validity	Predeclared endpoint/margin, valid sampling unit, sufficient resolution, multiplicity policy
G3	Baselines and ablations	Same inputs, dimensions, splits and evaluation budgets
G4	External and shift validity	Untouched data, complete datasheet, overlap audit, regime reporting
G5	Operational value	End-to-end cost, index type, accuracy/resource joint analysis
G6	CAL admissibility	Explicit outcome, metric-plane boundary, claim wording match
G7	Independent reproduction	Distinct environment/team, replay, discrepancy and review

Per-commit CI runs unit tests, numerical-gradient validation, contract cross-references, leakage guards and synthetic replay. Pull requests affecting the paper rebuild the PDF. Full datasets are not silently downloaded on every pull request. Qualification is a manual or protected workflow with pinned data, weights, environment and evidence destination.

8 MMALS replay and CAL

8.1 Immutable MMALS replay

The notebook is a client of the experiment specification, not the source of truth. Each run records resolved configuration, environment, dataset digests, implementation digest, events, audit decisions, route metrics, pair or search scores, thresholds, figures and artifact checksums. A compact projection supports activity replay while the full domain pack preserves scientific evidence.

Observed, derived and declared evidence are distinguished. Test opening, route freezing and gate evaluation are logical events. Replaying a bundle verifies its identity and decision history; it does not independently prove that the original population was representative.

8.2 CAL as a bounded claim gate

The internal MMALS–CAL contract has exactly three outcomes:

admissible all mandatory gates pass for the exact bounded statement;
inadmissible mandatory evidence contradicts the proposed statement;
indeterminate evidence is missing, incomplete, underpowered or out of scope.

The default is INDETERMINATE (machine token `INDETERMINATE`). It never silently becomes permission. Qualification reports coverage error, set size, calibration inflation, prediction bias, action and abstention rates, severity-weighted violations, false admissibility and GateRegret when those quantities are defined. They belong to the `decision_side_intervention` metric plane and cannot be relabelled as matcher FMR or FNMR. CAL conformance is an internal assurance property, not regulatory certification and not a complete safety proof.

9 Study 0: completed LFW audit

9.1 Purpose and data

Study 0 preserves the original educational setting to answer a narrow question: is the pair-supervised projection genuinely trained, and does it demonstrate non-inferiority under a stronger protocol? Frozen ImageNet ResNet-18 embeddings are extracted from the LFW development protocol. TRAIN contains 842 pairs, VALIDATION 795 pairs and untouched DevTest 1,000 pairs. TEST has 500 impostor pairs, so its empirical FMR step is 0.002. Five seeds were declared for every stochastic 128D route.

The extractor is not face-specific and the dataset cannot support industrial low-FMR claims. Those are limitations of the inference domain, not excuses to discard the observed result.

9.2 Predeclared decision

At target FMR $\alpha = 0.01$, the non-inferiority margin was $\delta = 0.03$ absolute FNMR. The method-level rule required every declared seed to have an upper paired-bootstrap bound at or below the margin. Validation-frozen thresholds were analysed separately.

9.3 Results

The Siamese route has mean $\Delta_{\mathrm{FNMR}} = 0.0228$ relative to raw, but the worst upper confidence bound across seeds is 0.156, above the declared margin. Neither Siamese, PCA nor random projection passes the

Table 3: Study 0 descriptive TEST equal-FMR benchmark at FMR 0.01.

Route	Dim.	FNMR mean	ROC AUC mean	EER mean
Raw	512	0.8060	0.7931	0.2890
PCA	128	0.8216	0.8058	0.2704
Random	128	0.8416	0.7726	0.3104
Siamese	128	0.8288	0.8167	0.2662

method-level non-inferiority rule. Claim `C-NI-001` is therefore NOT DEMONSTRATED. The higher mean AUC and lower mean EER of the Siamese route do not rescue the failed primary claim.

With thresholds selected on VALIDATION and frozen before TEST, mean (FNMR, FMR) is $(0.8940, 0.0020)$ for raw, $(0.8468, 0.0020)$ for PCA, $(0.8940, 0.0024)$ for random and $(0.9312, 0.0016)$ for Siamese. These values expose threshold-transfer behaviour; they are not an equal-FMR ranking.

9.4 Falsification outcome

Study 0 supports the mechanism claim: pair labels update a shared projection and produce consumable 128D templates. It does not support preservation, superiority or industrial value. The scientifically correct conclusion is "not demonstrated for this extractor, dataset and protocol", not "Siamese networks do not work" and not "compression works".

10 Revised priors and beliefs

Belief revision is recorded append-only in `beliefs/prior_posterior.yaml`. The current updates are:

1. belief that contrastive pair training updates the implemented projection increases;
2. belief that a simple linear 512-to-128 head naturally preserves low-FMR performance decreases for the Study 0 scope;
3. belief in metric learning in general remains unchanged because Study 0 cannot test that universal proposition;
4. belief in end-to-end engineering value remains unresolved because only payload arithmetic, not index or workload cost, has been measured.

Qualitative priors are acceptable during this exploratory phase. A confirmatory Bayesian claim would require quantitative priors and sensitivity analysis registered before data are opened.

11 Next registered gates

Study 1 remains DRAFT PREREGISTRATION, not "ready to run". Before execution it must pin the face-specific backbone and weights, licensing and provenance, target population, development and qualification data, overlap audit, primary FMR, non-inferiority margin, sample-size calculation, multiplicity policy, seeds and compute budget. Until then, CAL for `C-FACE-001` is INDETERMINATE.

The immediate engineering order is:

1. complete Study 1 preregistration without looking at its qualification results;
2. validate dataset acquisition and overlap tooling on non-qualification data;
3. execute the frozen 1:1 study and publish all routes and seeds;
4. decide from the registered gates whether compression ablation is justified;
5. only then add 1:N indexes and constrained-device benchmarks;
6. reproduce the strongest bounded result independently before any broad claim.

12 Threats, ethics and limits

Face embeddings are biometric data. Dataset access, processing, retention and publication must follow applicable authorisation and licences. Raw images and recoverable templates are not redistributed by this repository. Hashes, schemas and derived aggregate evidence are not a substitute for privacy and security review.

Performance differences can arise from image quality, capture conditions and population composition. Aggregate performance can hide regime-specific failure. When authorised data support it, Study 3 reports predeclared regime results and uncertainty. Absence of such data is reported as a limitation, not interpreted as parity.

Presentation-attack detection, adversarial robustness, template protection and regulatory certification are outside the current claim scope. A smaller unprotected embedding is not automatically a safer template.

13 Reproducibility statement

The repository contains source code, configurations, notebooks, tests, machine-readable claims, dataset requirements, gates and Study 0 results. The run identifier, configuration digest, implementation digest and artifact hashes are recorded. LFW images and cached face features are excluded. A smoke run checks numerical and evidence plumbing but is explicitly forbidden from supporting a biometric claim.

The paper is rebuilt from versioned source. CI checks required traceability tokens and cross-file references. Future work should generate result tables directly from qualified replay bundles, eliminating manual duplication present in this protocol draft.

14 Conclusion

Siamese metric learning is neither accepted nor rejected as a slogan. The research object is a versioned compression intervention evaluated against explicit alternatives at decision- relevant operating points and under measurable system constraints. Study 0 confirms that the training mechanism is real and falsifies the proposed non-inferiority claim under its limited setting. The programme now makes the harder questions executable: whether a face-specific source representation changes that outcome, whether supervision beats ordinary compression, whether thresholds transfer, whether 1:N search benefits, and whether the resulting claim is admissible under replayable evidence. The value of the programme is not that every gate will pass; it is that failures remain visible, bounded and informative.

A Machine-readable evidence map

Path	Role
`protocol/research_program.yaml`	Umbrella scientific object and estimand
`protocol/studies/*.yaml`	Study status, preregistration, run and decision
`claims/registry.yaml`	Claim wording, status, evidence and required gates
`datasets/*.yaml`	Dataset evidence and qualification requirements
`gates/gate_spec.yaml`	G0–G7 qualification contract
`gates/cal_spec.yaml`	Three-state internal admissibility contract
`beliefs/prior_posterior.yaml`	Append-only belief updates
`run_manifest.json`	Immutable run identity, environment and checksums
`data/events.jsonl`	Logical execution history
`audit_trace.jsonl`	Observed, derived and declared decisions

B References

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